

CUSTOMER SEGMENTATION & CLV DASHBOARD

Power BI case study using real online retail transaction data

397,884

cleaned transaction rows

4,338

customers analyzed

64.6%

champion revenue share

2 pages

overview + drillthrough

Python + BI

data prep and dashboard

Customer Segmentation & CLV Dashboard

Real online retail transaction data | Customer value, behavior, and retention analysis

8.91M

Total Revenue

64.59%

Champion Revenue %

4.34K

Total Customers

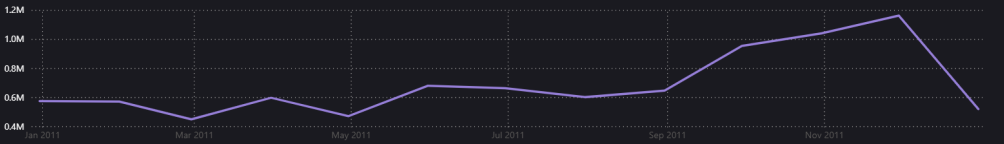
2K

At-Risk Customers

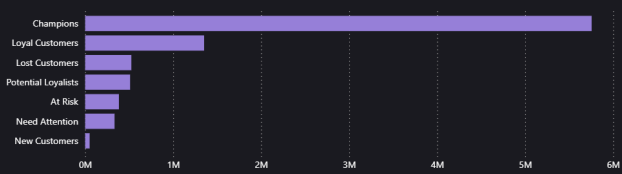
€ 2.95K

Avg Annual CLV

Monthly Revenue Trend



Revenue Concentration by Segment



Top Customers by Revenue

CustomerID	Segment	Revenue	Orders	AVG Annual CLV
14646	Champions	280,206.02	73	288,099.15
18102	Champions	250,657.30	60	258,242.27
17450	Champions	194,550.79	46	192,964.78
16446	Potential Loyalists	168,472.50	2	298,507.10
14911	Champions	143,825.06	201	140,748.34
12415	Champions	124,914.53	21	134,892.91
14156	Champions	117,379.63	55	115,170.87
17511	Champions	91,062.38	31	88,871.04
16029	Loyal Customers	81,024.84	63	79,075.04
17346	Lost Customers	77,103.68	1	86,417.77

Customer Drillthrough Profile

Real online retail transaction data | Customer value, behavior, and retention analysis

16.45K

Selected Customer ID

168.47K

Customer Revenue

84.24K

AVG Order Value

€ 298.51K

Average Annual CLV

1

Recency days

Promising customer with recent activity. Encourage repeat purchases through bundles, reminders, and targeted discounts.
Customer Action Detail

Convert to Loyal Customer
Customer Action Recommendation

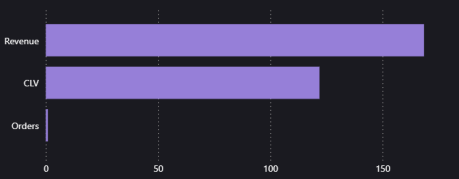
Potential Loyalists

Selected Segment

Products Purchased by Selected Customer

Description	Quantity	Orders	Revenue
PAPER CRAFT - LITTLE BIRDIE	80995	1	168,469.60
PANTRY SCRUBBING BRUSH	1	1	1.65
PANTRY PASTRY BRUSH	1	1	1.25

Customer vs Segment Benchmark



Project Overview

Business problem

- A retail business needs to understand which customers drive value.
- Customer revenue is not evenly distributed across the customer base.
- The business needs a practical way to identify high-value, loyal, at-risk, and reactivation candidates.

Project objective

- Build a Power BI dashboard that segments customers using RFM logic.
- Estimate customer value using cleaned customer purchase behavior.
- Create an interactive drillthrough page that turns analysis into customer-level action.

Dashboard structure

01 Customer Intelligence Overview

Revenue, customer base, champion revenue share, risk customers, segment concentration, monthly trend, and top customer table.

02 Customer Drillthrough Profile

A selected customer view showing revenue, CLV, recency, recommendation, segment benchmark, and purchased products.

Data Preparation Methodology

Source data

- Dataset: UCI Machine Learning Repository - Online Retail.
- Contains invoice-line transactions for a UK non-store online retail business.
- Raw file fields include InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

Cleaning logic

- Removed rows with missing CustomerID to keep customer-level analysis reliable.
- Removed cancelled invoices beginning with C.
- Removed negative or zero quantity and unit price records.
- Converted CustomerID to text and InvoiceDate to datetime.
- Created TotalSales as Quantity multiplied by UnitPrice.

Customer modeling

- Calculated Recency, Frequency, and Monetary value for every customer.
- Generated R, F, and M scores using quintile ranking.
- Assigned customer segments such as Champions, Loyal Customers, Potential Loyalists, At Risk, Lost Customers, and Need Attention.
- Estimated annual CLV using customer age from first purchase to dataset snapshot date to avoid inflated one-day customer values.

Dashboard Page 1 - Customer Intelligence Overview

Customer Segmentation & CLV Dashboard

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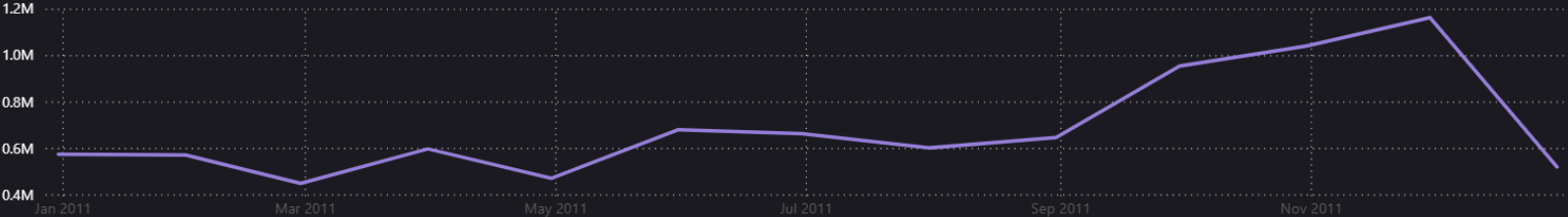
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At-Risk Customers

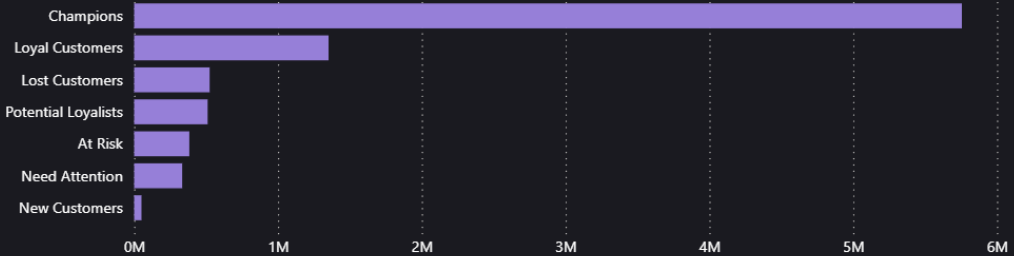
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Page 1 Walkthrough

What this page answers

- How much revenue was generated by the cleaned transaction dataset?
- How many customers are available for segmentation?
- How concentrated is revenue inside the Champion customer segment?
- How many customers fall into risk-related segments?
- Which customers should be investigated through drillthrough?

Visual logic

KPI row

Shows total revenue, customer count, champion revenue share, at-risk customers, and average annual CLV.

Monthly Revenue Trend

Acts as the page hero visual and shows revenue movement across 2011.

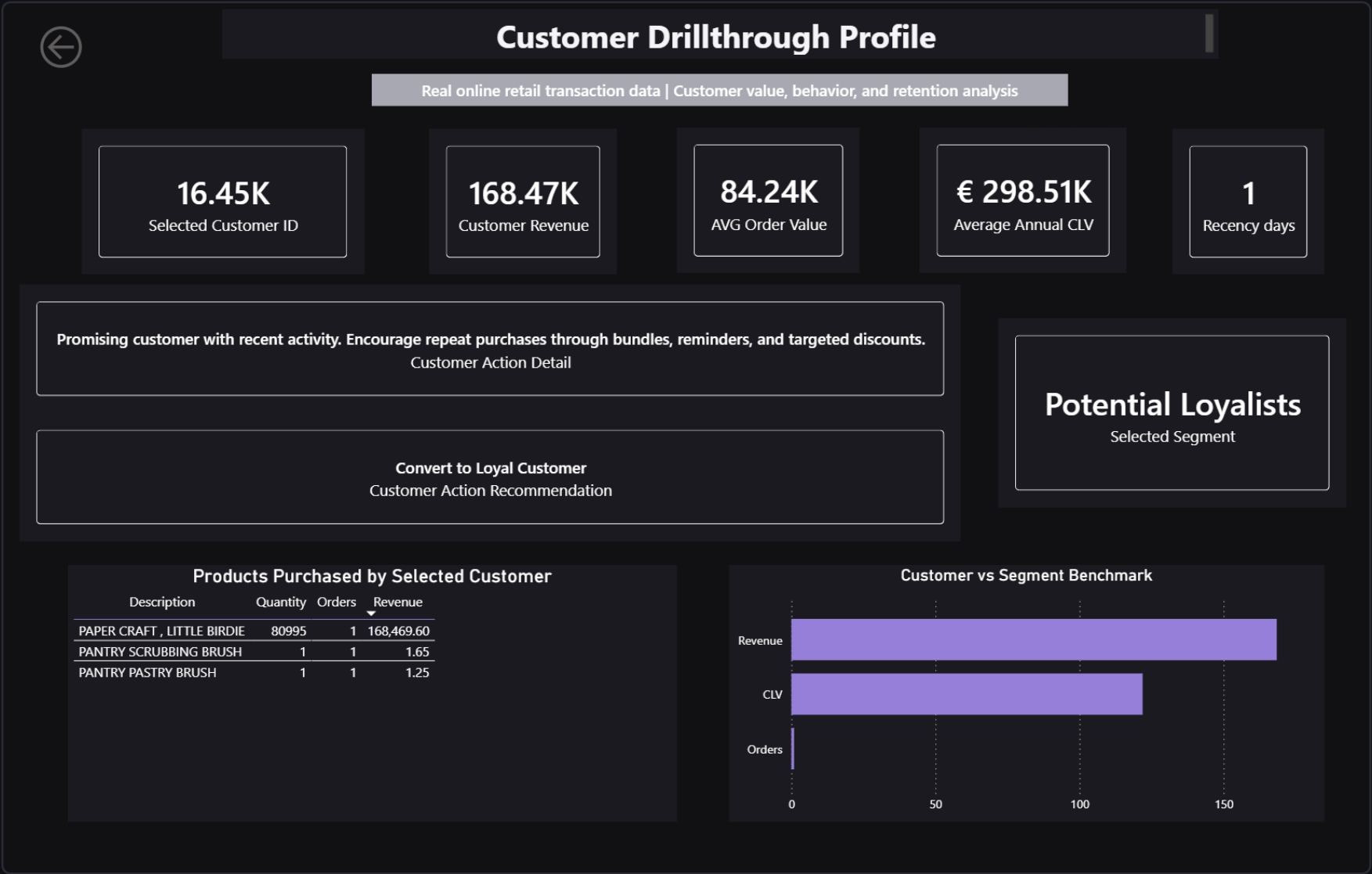
Revenue Concentration by Segment

Shows Champions dominate revenue, making retention and loyalty strategy important.

Top Customers by Revenue

Provides the customer-level entry point for drillthrough analysis.

Dashboard Page 2 - Customer Drillthrough Profile



Page 2 Walkthrough

Drillthrough behavior

- The user right-clicks a customer from Page 1 and drills through into the Customer Profile page.
- The page filters to the selected customer and displays customer-specific revenue, order value, CLV, recency, products, and segment benchmark.
- In the screenshot, Power BI uses compact formatting for the selected customer ID card, but the underlying drillthrough filter still targets the selected customer record.

Decision logic

- The recommendation card translates the selected customer segment into a suggested business action.
- Potential Loyalists receive repeat-purchase prompts such as bundles, reminders, and targeted discounts.
- The benchmark chart compares the selected customer against the average customer in the same segment as a multiple.
- The product table shows which products the selected customer actually purchased, helping the business personalize follow-up offers.

Technical Summary & Portfolio Value

Tools

- Python
- pandas
- Power BI
- DAX
- Power Query
- GitHub

Core outputs

- cleaned_transactions.csv
- customer_rfm.csv
- segment_summary.csv
- monthly_revenue.csv
- Power BI dashboard

Skills shown

- Data cleaning
- Customer segmentation
- RFM modeling
- CLV estimation
- DAX measures
- Drillthrough UX
- Business recommendations

Conclusion

This project demonstrates a full analytics workflow: real transactional data was cleaned in Python, transformed into customer-level RFM and CLV outputs, modeled in Power BI, and presented through an interactive dashboard with customer-level drillthrough. The result is not just a static report; it is a practical customer intelligence tool for identifying high-value customers, risk groups, and retention actions.